

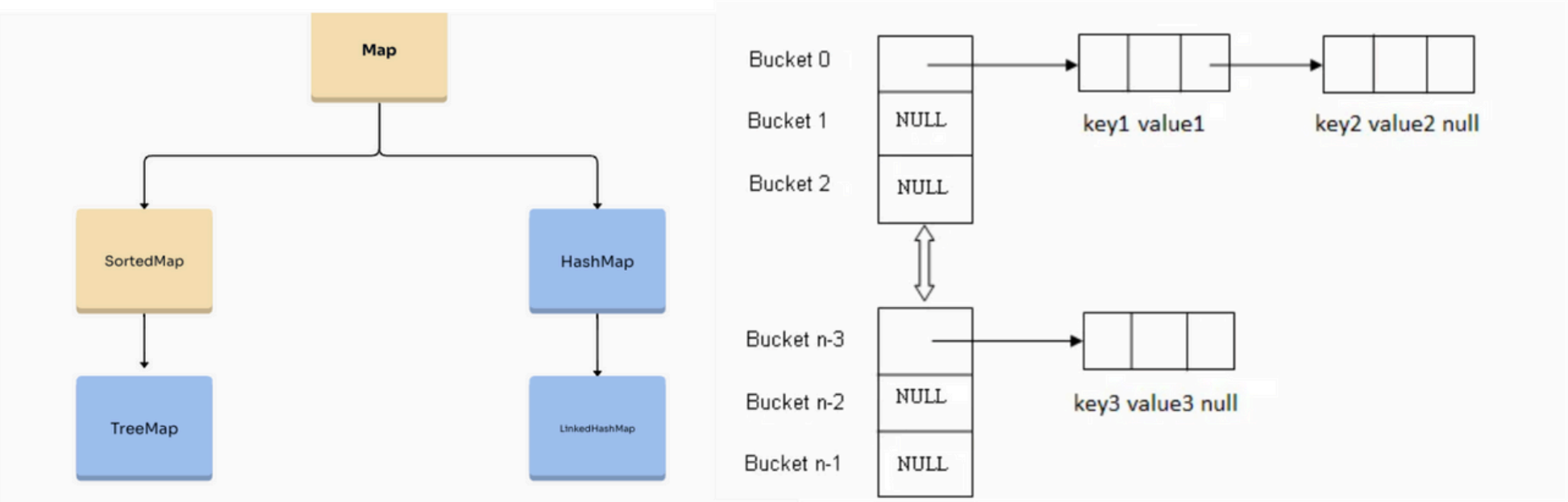


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HOW HASHMAP WORKS INTERNALLY

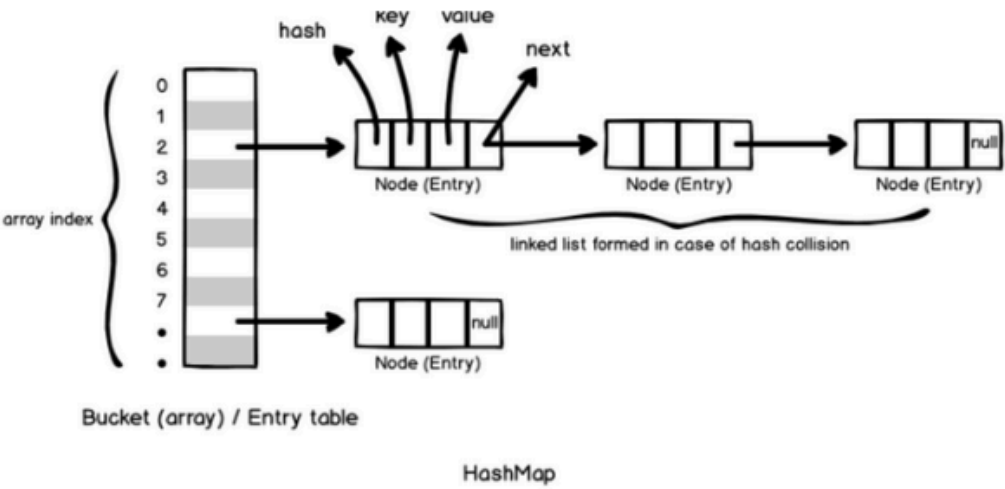
What is Hashmap



- Technically, HashMap is one of the implementation classes of the Map interface.
- It is used to store data in the form of key-value pairs.
- Here, the key must be unique, and the value can be duplicated.

HashMap Base Data structure

```
public class HashMap<K, V> implements Cloneable, Serializable {  
    // Inner Class Node  
    static class Node<K, V> {  
        final int hash;  
        final K key;  
        V value;  
        Node<K, V> next;  
    }  
  
    // Inner Interface Entry  
    interface Entry<K, V> {  
        K getKey();  
        V getValue();  
        V setValue(V value);  
    }  
  
    // HashMap functionality and other inner classes  
}
```



HashMap Internal Structure

- HashMap is implemented based on a **Hash Table data structure**.
- The Hash Table itself is implemented using an **array and a linked list data structure**.
- When I create a HashMap, it will internally create 16 buckets or nodes by default.
- Each bucket in the HashMap is implemented using a Linked List data structure - These buckets are **Linked List nodes**.



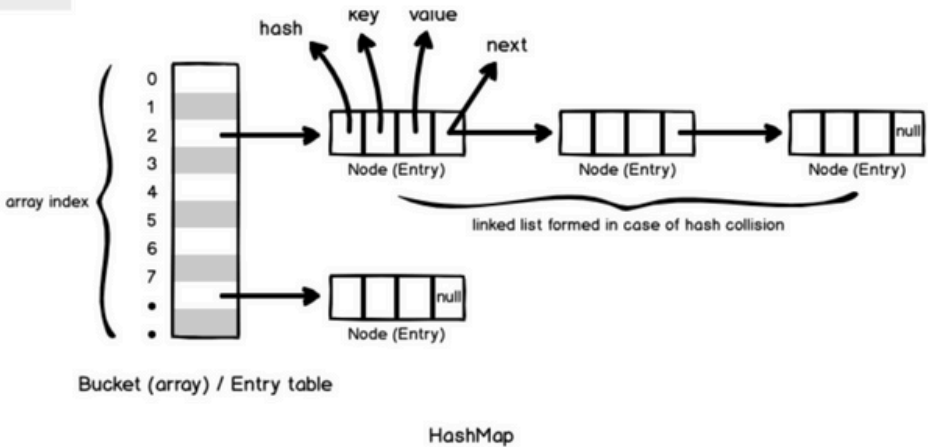
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HOW HASHMAP WORKS INTERNALLY

Role of Node Class & Entry Interface in the HashMap Class

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        K getKey();  
        V getValue();  
        V setValue(V value);  
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    // HashMap functionality and other inner classes  
}
```



- HashMap has an inner class called Node.
- Each node contains 4 pieces of information
 - #1: Hashcode of the Key (a unique hash number used as an index)
 - #2: Key
 - #3: Value
 - #4: Address of the next node.
- HashMap stores all entries into buckets using a hashing technique.
- It also has an inner interface called Entry - Used access, modify, and iterate over individual entries in the map.

How the put() operation works internally.

What is put() method

- The put() method is used to add entries to a HashMap.
- Here, entries mean key-value pairs.



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HOW HASHMAP WORKS INTERNALLY

How the put() operation works internally.

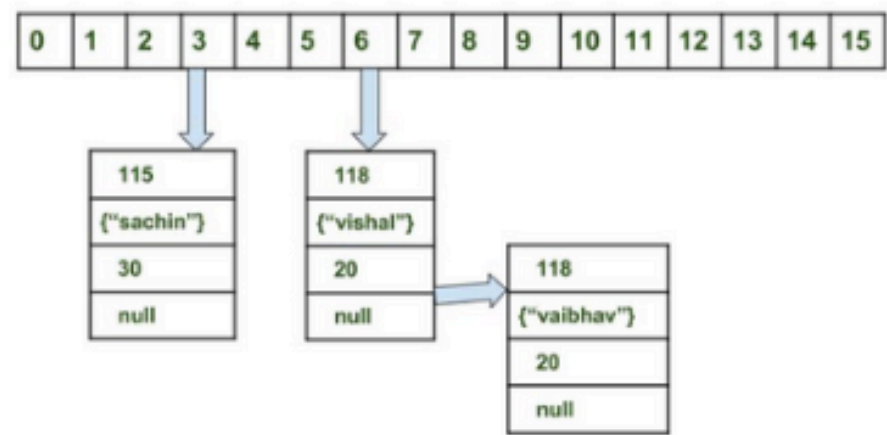
Case 1: Add a new entry using the put() method



- Whenever we try to add a new entry using the put() method
- The put() method calculates a hash code for the key.
- Internally, the put() method calls the hash() function to calculate the hash code for the key.
- The hash() method calculates the hash number for the key by applying some modular logic.
- Once the put() method gets a hash number for the key, it stores the entry in a bucket based on this hash number.
- In each bucket, entries are stored in the form of a linked list data structure.
- Internally each entry contains four pieces of information
 - #1: HashCode of the key (calculated hash number used as the index for the entry)
 - #2. Key
 - #3. Value
 - #4. Address of the next node

How the put() operation works internally.

Case 2: What if two entries have the same hashcode



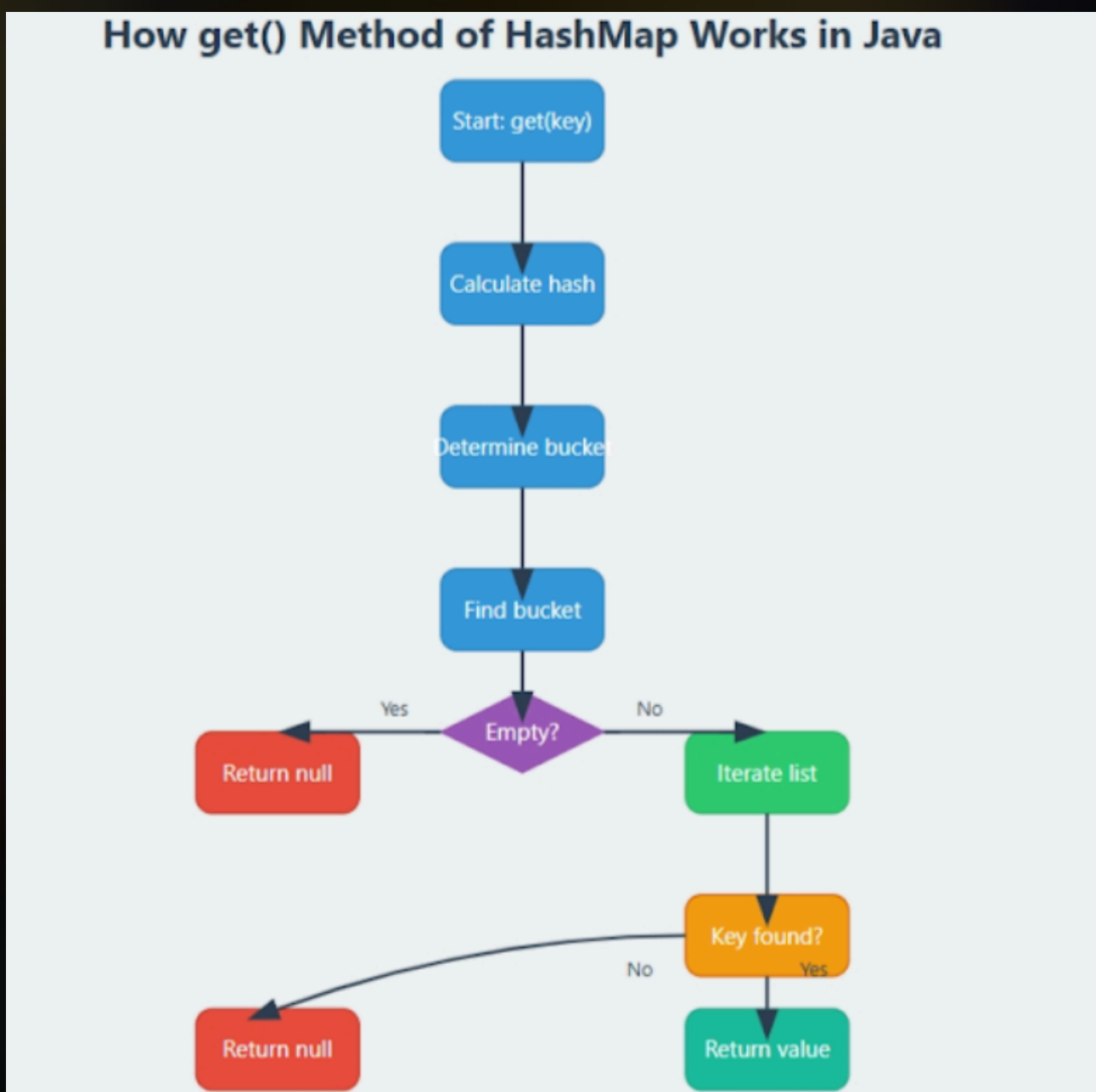
- If two entries have the same hash code, then both entries will be stored in the same bucket.
- Here, the existing node will have a reference to the new node.
- Internally, HashMap uses a concept called “separate chaining technique” to handle collisions.
- So, even if multiple entries have the same hash code, they will be stored in separate nodes within the linked list of the same bucket.
- A new node is always added at the end of the linked list.



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How Hashmap Works internally



How the get() operation works internally.

What is get() method

- get () method is used to retrieve a value based on the key
- get () method converts key into hashNumber using hash () method
- hash () function uses modular operation to calculate hash number of a key
- Based on the hash number in the bucket it will retrieve corresponding value



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PURPOSE OF THIS DOCUMENT

- **Use this PDF as a quick last-minute revision guide only.**
- **For more details, please perform R&D and gain a deeper understanding on each topic.**
- **Once the interview series is complete, we will cover this topic in detail.**



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